

# Topic: Line Integrals

## I. Definition & Properties

Defn: A **Line Integral** is integration over a curve,  $C$  is given by

Vector-valued functions:

Scalar-valued functions:

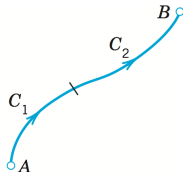
Note: The expressions above are valid whether  $C$  is a planar or space curve.

Properties: Let  $\mathbf{F}$  be vector-valued function and  $k$  a constant.

$$\circ \int_C k \mathbf{F} \cdot d\mathbf{r} =$$

$$\circ \int_C (\mathbf{F} + \mathbf{G}) \cdot d\mathbf{r} =$$

$$\circ \int_C \mathbf{F} \cdot d\mathbf{r} =$$



Ex: If  $\mathbf{F} = [y^2, -x^2]$  and  $C : y = 4x^2$  from  $(0, 0)$  to  $(1, 4)$ , then compute  $\int_C \mathbf{F} \cdot d\mathbf{r}$

Soln:

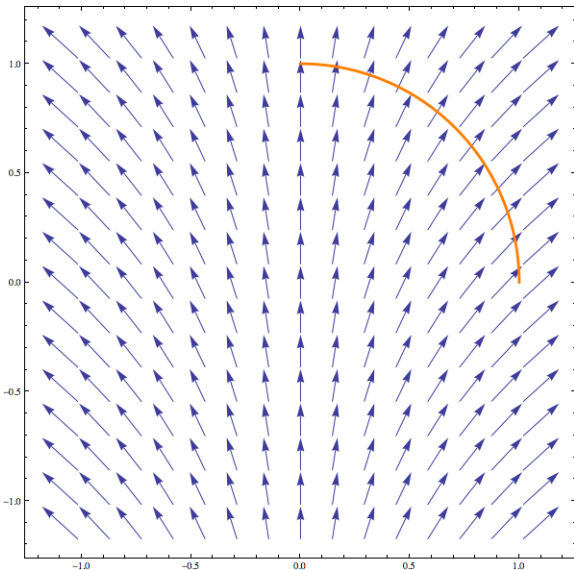
Ex: Determine  $\int_C \mathbf{F} \cdot d\mathbf{r}$  where  $\mathbf{F} = [x, y - z^2, z - x]$  and  $C : \mathbf{r}(t) = [2 \cos t, t, 2 \sin t]$  from

$(2, 0, 0)$  to  $(2, 2\pi, 0)$

Soln:

## II. Work

Suppose there was a mass moving through a planar force field,  $\mathbf{F}$ , along a path,  $C$ .



Force field:

$$\mathbf{F}(x, y) = [x, 1]$$

Parameterization of C:

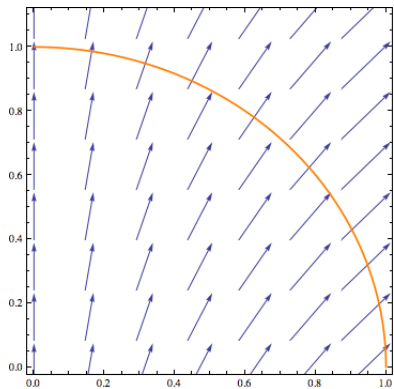
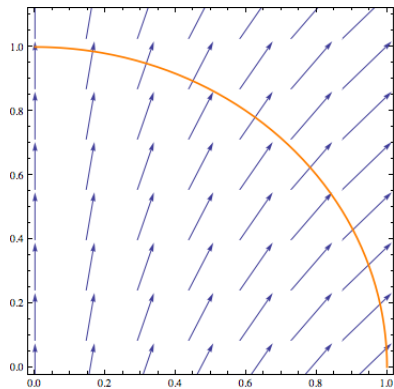
$$\mathbf{r}(t) = [\cos t, \sin t], \quad 0 \leq t \leq \frac{\pi}{2}$$

What is the work done in moving this mass along C?

Previously, we calculated work using the dot product:

$$W = \mathbf{F} \cdot \mathbf{r}$$

However, since both the force and displacement change at every point along C, we are forced to add up “many” of “small works” at different points (i.e. integrate).



"Small" Work =

$$dW =$$

Total Work =

$$W =$$

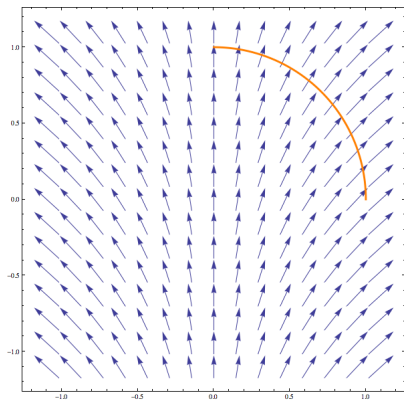
Ex:  $\mathbf{F} = [x, 1]$  with  $\mathbf{r}(t) = [\cos t, \sin t]$ ,  $0 \leq t \leq \frac{\pi}{2}$

What is the work done to move the object along this path when exposed to  $\mathbf{F}$ ?

$$\mathbf{F}(\mathbf{r}(t)) =$$

$$\mathbf{r}'(t) =$$

$$\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) =$$



Ex: Let  $\mathbf{F} = [xy, 2y^2]$  represent a force vector field in the x-y plane. If a particle moves along a quarter circle from  $(2, 0)$  to  $(0, 2)$ , then find the work done on that particle.

Want: \_\_\_\_\_, Need:

$$\mathbf{F}(\mathbf{r}(t)) =$$

$$\mathbf{r}'(t) =$$

$$\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) =$$

Ex: Suppose a wire has density  $\delta(x, y, z) = 3\sqrt{5+x}$ . Thinking of the wire as a curve in space, it could be parameterized by  $\mathbf{r}(t) = t\mathbf{i} + 2t\mathbf{j} + \frac{2}{3}t^{3/2}\mathbf{k}$  where  $0 \leq t \leq 2$ . Find the mass of the wire.

Note 1:

Note 2:

$\Rightarrow$     *Mass* =